MIT-WPU | Faculty of Engineering and Technology

School of Computer Science and Engineering

Department of Computer Engineering and Technology

T.Y. B.Tech CSE **AY 2026-27** **Semester 5**

Computer Networks Lab (CSE20110)

Assignment No.: 5

Title: Write a program for TCP Socket Programming: (Language: C/C++/Python/Java).

Aim: Write a C program for wired network using TCP socket to demonstrate

a. Chat application (50% students)
b. Mathematical operations. (Remaining 50% students)

Objectives:

1. To understand the concept of socket programming.

Theory: (Handwritten & Scanned)

Students should write following theory

1. What is Socket?
2. TCP Socket Primitives at Server and Client

Pseudocode:

Server Program:

1. START
2. Create TCP socket
3. Bind socket to IP address and port number
4. Listen for client connection
5. Accept client connection
6. Receive message from client
7. Display received message
8. Send reply to client
9. Close client connection
10. Close server socket
11. STOP

Client Program:

1. START
2. Create TCP socket
3. Connect to server IP address and port number
4. Input message from user
5. Send message to server
6. Receive reply from server
7. Display server reply
8. Close client socket
9. STOP

Program Code :

- Students should write/paste code here

Output Screenshots:

- Students should Draw/Paste final output screenshots

Students Observation:

- Students should write their observations about lab assignment

FAQ (Handwritten & Scanned)

1) What is a TCP Socket?
2) What are the main steps involved in TCP Socket Programming?
3) What is the role of IP address and port number in TCP Socket Programming?
4) Why is TCP called a connection-oriented protocol?
5) What is the difference between TCP and UDP socket primitives?